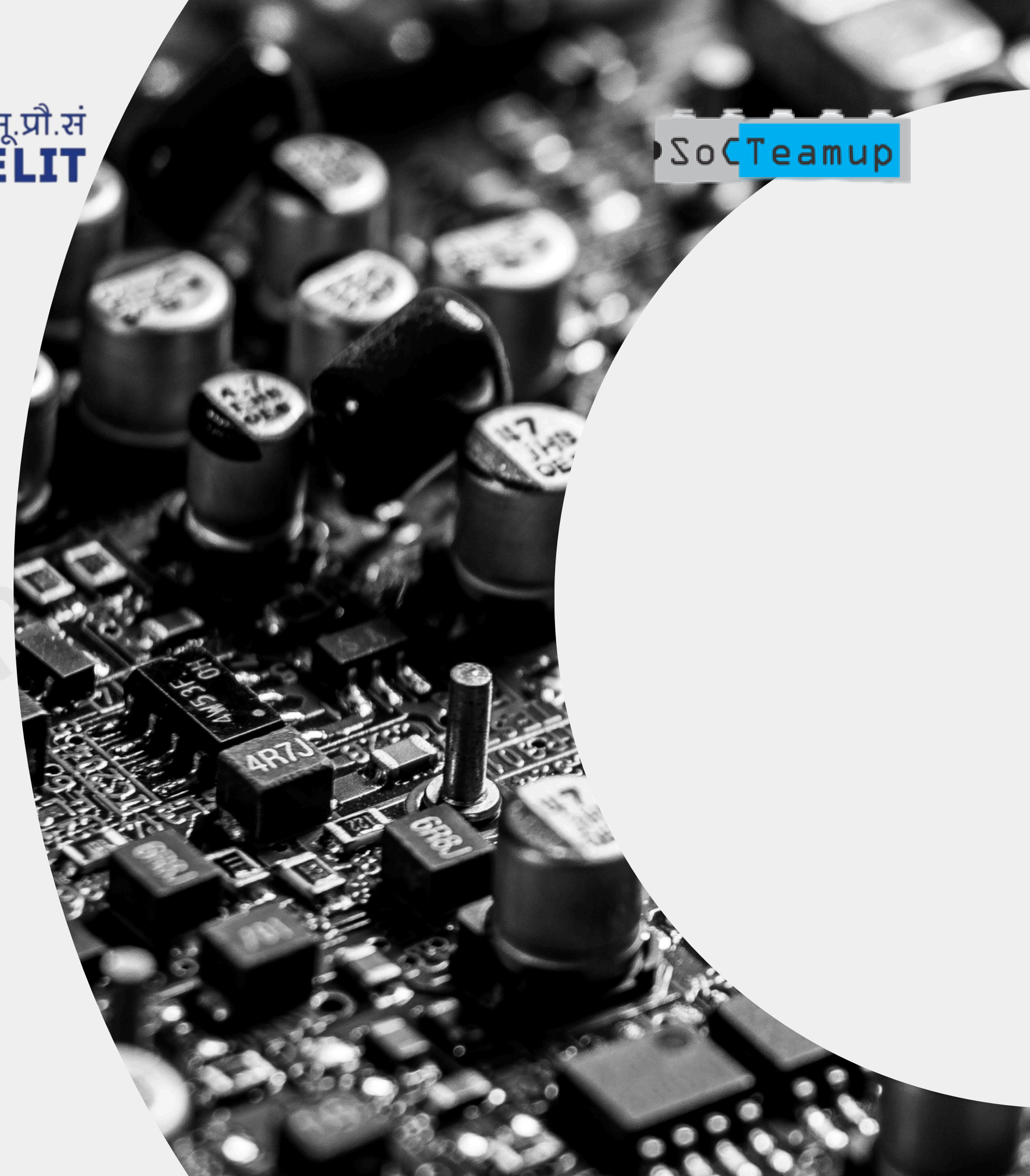


SoC Clocking and Reset Architecture





Outline

Role of Clock and Reset in SoC Operation

Clock Tree Structure and Distribution

PLLs, Dividers, and Clock Muxing

Gated Clock Architecture

Reset Tree Design and Propagation

Reset Generation and Synchronization

Summary & Industry Design Guidelines

➔ Role of Clock and Reset in SoC Operation



CLOCK:

- DRIVES TIMING OF SEQUENTIAL LOGIC
- DEFINES DATA FLOW AND OPERATION RATE.



RESET:

- INITIALIZES STATE MACHINES AND REGISTERS
- REQUIRED FOR SAFE BOOT-UP AND TESTABILITY

Role of Clock and Reset in SoC Operation

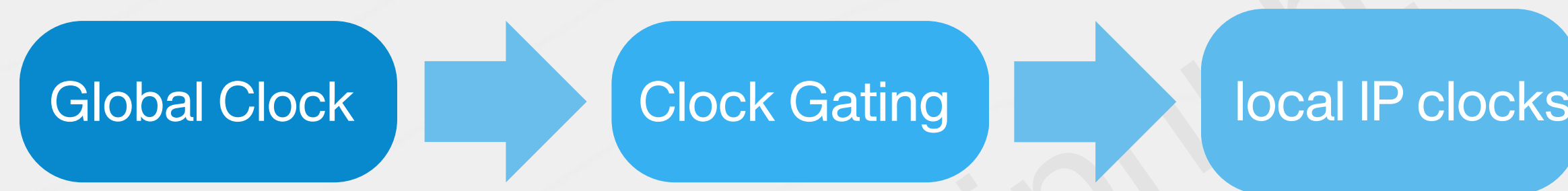


IMPACT:

- ✓ POWER, PERFORMANCE, AREA (PPA)
- ✓ CLOCK DOMAIN PLANNING AND SKEW CONTROL
- ✓ CLEAN RESET ENSURES PREDICTABLE SOC BEHAVIOUR

➔ Clock Tree Structure and Distribution

Clock Tree Overview:



Types:	Clock Tree Synthesis (CTS):
• Global Clock Network • Leaf Clock Buffers • Skew Management Buffers	• Balances delay paths • Controls clock skew • Involves physical-aware placement

PLLs, Dividers, and Clock Muxing

- **PLL (Phase-Locked Loop):**
 - Generates high-speed clocks
 - Can output multiple frequencies from reference
- **Clock Divider:**
 - Reduces frequency (e.g., $\text{sys_clk}/2$)
- **Clock Mux:**
 - Selects clock source (PLL or backup)
- **Design Note:**
 - Use glitch-free muxes with proper enable logic
 - Document source-to-sink path for each domain

Gated Clock Architecture

- **Why Gate Clocks?**
 - Save dynamic power by stopping inactive logic
- **Clock Gating Styles:**
 - RTL clock gating (using enable)
 - Clock gating cells (inserted during synthesis)
 - Integrated clock gating (ICG cells inputs)
- **Best Practices:**
 - Gate as close to the leaf as possible
 - Use synthesis DFT-aware tools

Reset Tree Design and Propagation

- **Reset Tree:**
 - Distributes resets from SoC-level controller
- **Types of Resets:**
 - Power-on Reset
 - Warm Reset
 - Debug Reset
 - Block-level Reset
- **Planning Considerations:**
 - Align reset tree with clock tree
 - Avoid combinational loops in reset path
 - Sync de-assertion with receiving clock

Summary – SoC Clock & Reset Guidelines

- Centralize clock and reset planning early
- Minimize skew and latency with CTS and synthesis
- Use PLLs and dividers for flexibility
- Avoid glitches with muxes and synchronizers
- Gate clocks safely to reduce power
- Document and simulate all reset paths

Outcome:

- Robust, low-power, silicon-ready clock-reset strategy

Thank You

echiphu

